



Metropolitan Area (Wireless) Network

Installation / Support Manual

For

Solomon Islands Government

Issue 1



30 July 2026

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2 Document details

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0.1	21 th October 2012	Document draft	Carmelo Moschella
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0.3	00 November 2012	Commercial review	David Sexton
0.4	00 November 2012	Quality assurance	Mik Madden
1.0	00 November 2012	Document submission	David Sexton

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5 Disclaimer

Accuracy

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6 Installation – Base Station Unit Sites

This section provides necessary information unique to a Base Station Unit site. The following pages of notes refer to the Appendix pages which list instructions pertaining to the site. Photocopy the pages as required, and then take them on site as a reference for correct installation:

6.1 Site ID: SATSOL-TX-BSU-001/ SatSol TX

Site Information

Site ID: SATSOL-TX-BSU-001

Site Name: SatSol TX

Build Category: D and A

Facing Site ID: GOV-POL-BSU-002

Antenna Tilt: 235.77°

Bearing: 2.9°

Facing Site ID: SI-NDMO-BSU-003

Antenna Tilt: 2.0°

Bearing: 251.09°

Facing Site ID: GOV-GG-BSU-005

Antenna Tilt: 2.0°

Bearing: 265.08°

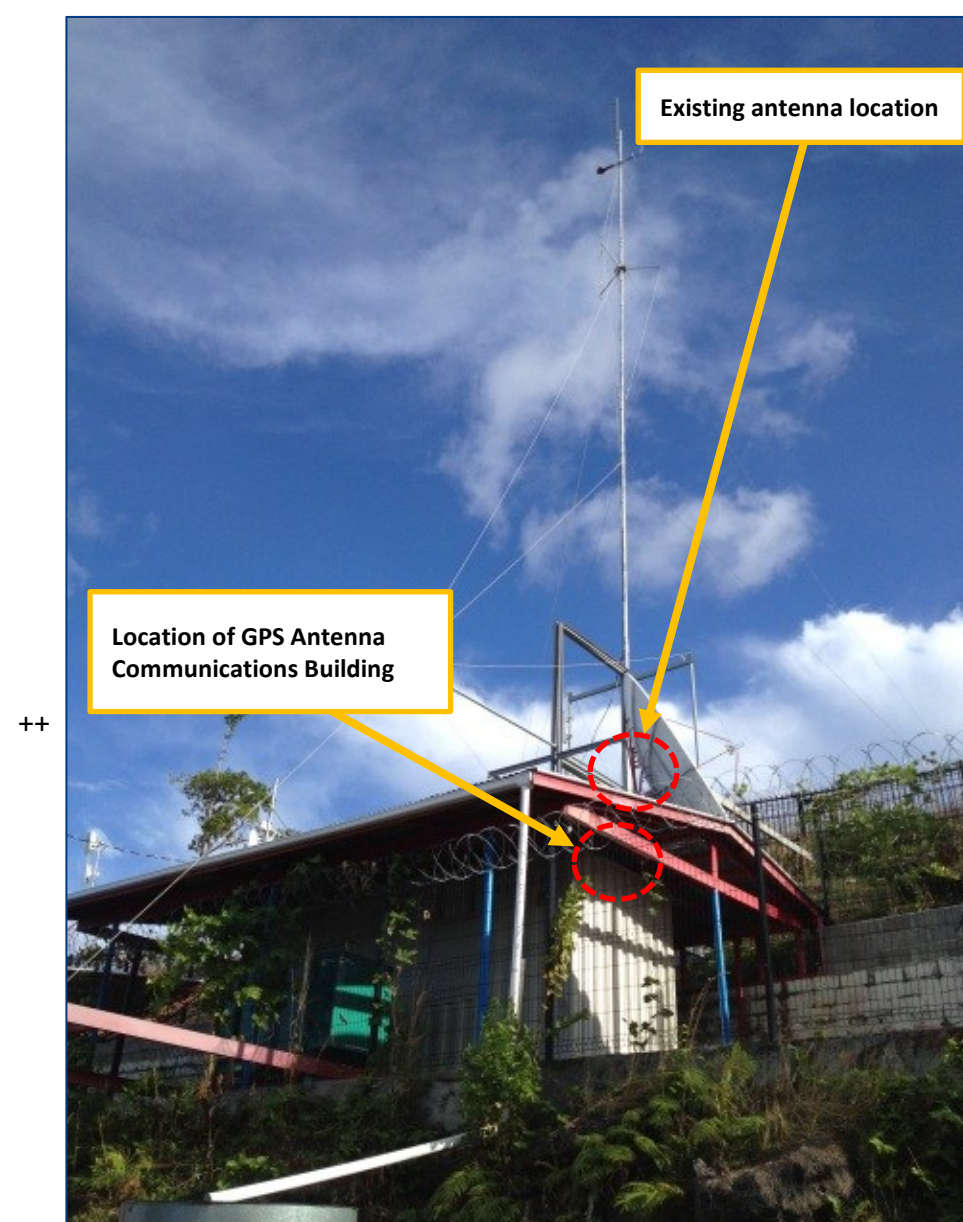


Figure 1

INSTALLATION NOTES

MAST INSTALLATION ---See Appendix B / Category A --Hockey Stick, Category D --30 ft. Mast

CABLING ---See Appendix C / Section 11.1

NETWORKING (RACK) Internal---See Appendix E /Category A /Section 13.1.1

BUILDING ENTRY POINTS---See Appendix F / Section 14.1

LIGHTNING PROTECTION UNIT ---- See Appendix G / Section 15.1

EARTHING See Appendix H / Section 16.1- General/ 16.2-PTP-PMP / Section 16.3-Tower/ 16.4-PMP

ANTENNA ASSEMBLY ---- See Appendix I / Section 17.1/17.2/17.3/17.4

WIMAX INSTALL ---- See Appendix J / Section 18.1

MICROWAVE ALIGNMENT ---- See Appendix K / Section 19.1

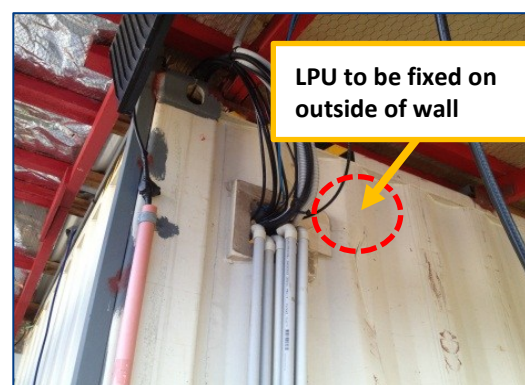


Figure 2

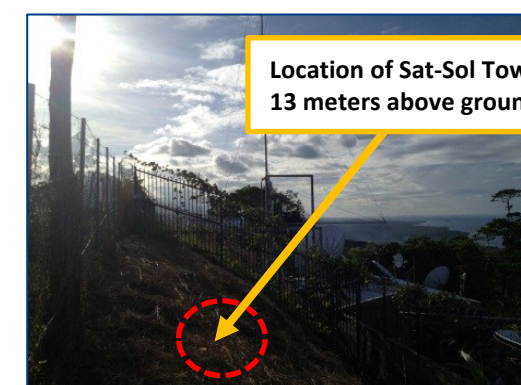


Figure 3

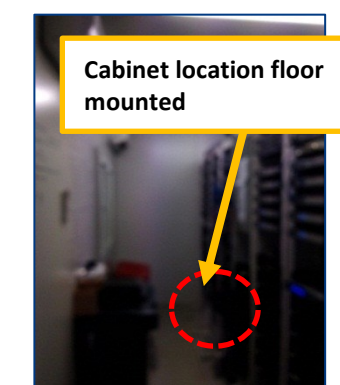


Figure 4

ROOF INSULATION NOTES

Cable will be installed between building and Tower utilising installed 100mm underground conduit. A cable snake will be used to pull cables through from one side to the other.

On the tower side, cables will be cable tied externally up the side of the tower using cable clips which will need to be installed every 500mm to support the cables. Each cable will break out to its respective networking equipment as it reaches its mount location.

At the building end cables will exit the 100mm underground conduit via 32mm corrugated conduit that will terminate at an IP66 rated enclosure mounted to the wall of the building. Inside the IP66 enclosure LPUs for each radio will be installed of which each cable entering the box will terminate to a designated location. Each LPU will have a second cable terminated in the second port of the unit which will exit the IP66 enclosure via a 32mm corrugated conduit which will be installed to the appropriate building entry point. Once cables are inside it will be run to and terminated at its designated cabinet location.

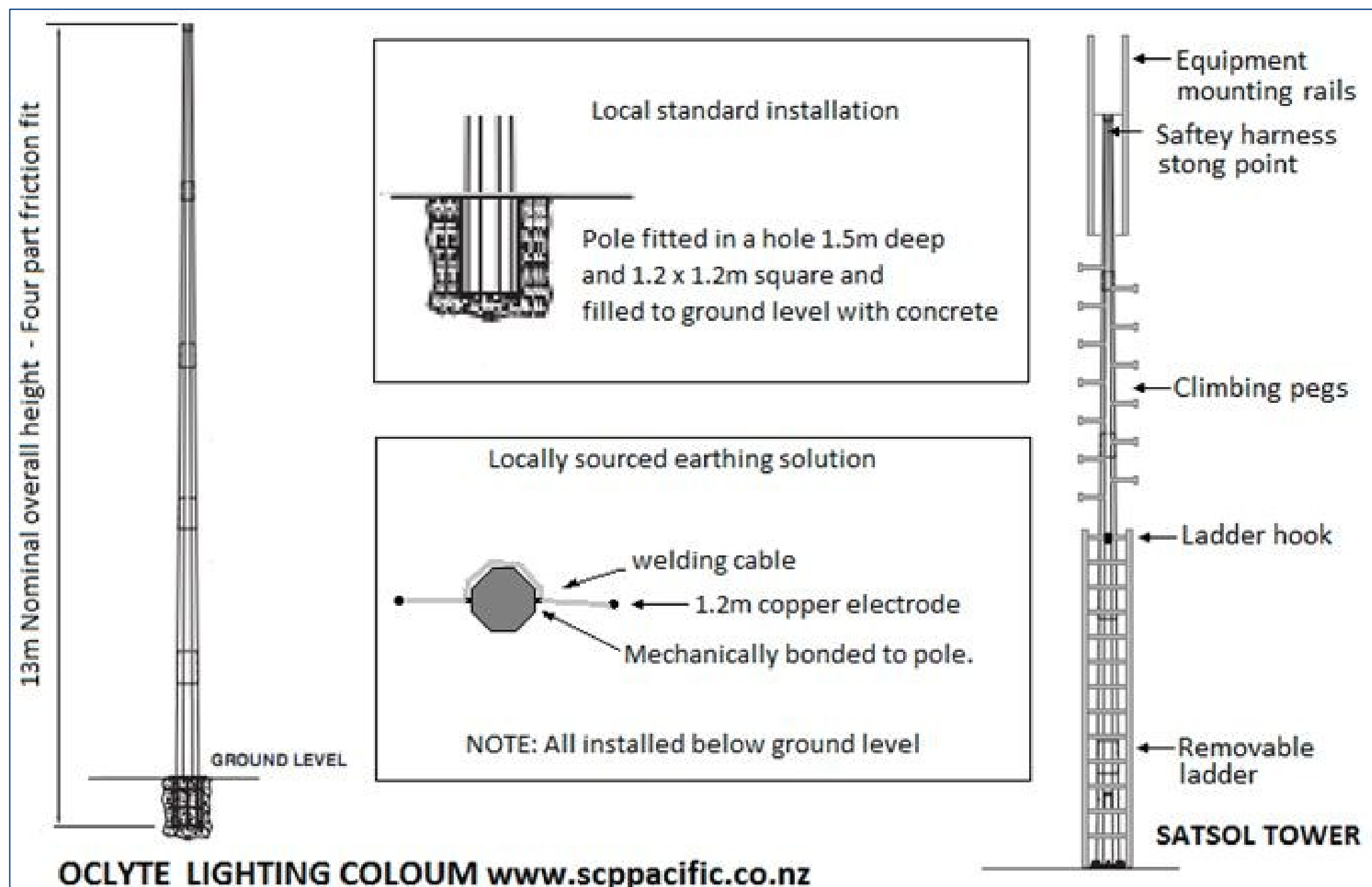


Figure 5 – Footing Design for Tower

6.1.1 Configuration Table

BSU PTP LINK Details (Backhaul locations)

BSU locations	Master Location	Slave Location	Master Building	Bearing	Antenna Tilt	Install Category	Slave Building	Bearing	Antenna Tilt	Install Category
LINK B.1	SATSOL-TX-BSU-001	GOV-POL-BSU-002	SatSol TX	55.78°	-2.9°	Category C	GOV-POL-BSU-002	235.77°	2.9°	Category D

BSU LOCATIONS AND HARDWARE

Site	Location ID	GPS	Hardware ID	IP Address	Hardware	Antenna ID	Azimuth
SATSOL-TX-BSU-001	09:26:50.4S, 159:55:03.6E	09:26:50.4S, 159:55:03.6E	BSU-TX-001	172.16.0.10	Cambium PMP 36320	PCTEL SP3338-16XP90	20.14

HT PTP LINK Details (Backhaul to HTP)

Link ID	Location ID	Hardware ID	Building	IP Address	Hardware
SatSol TX	SATSOL-TX-BSU-001	SATSOL-TX-SW-001	SatSol TX	172.16.2.100	WS-C3560CG-8PC-S

BSU PTP LINK Details (Backhaul locations)

BSU locations	Master Location	Slave Location	Master Building	Bearing	Antenna Tilt	Install Category	Slave Building	Bearing	Antenna Tilt	Install Category
LINK B.2	SATSOL-TX-BSU-001	SI-NDMO-BSU-003	SatSol TX	71.10°	-2.0°	Category C	NDMO Tower	251.09°	2.9°	Category D

BSU LOCATIONS AND HARDWARE

Site	Location ID	GPS	Hardware ID	IP Address	Hardware	Antenna ID	Azimuth
SATSOL-TX-BSU-001	09:26:50.4S, 159:55:03.6E	09:26:50.4S, 159:55:03.6E	BSU-TX-002	172.16.0.11	Cambium PMP 36320	PCTEL SP3338-16XP90	82.08

Additional Hardware Details

Site	Location ID	GPS	Hardware ID	IP Address	Hardware	Antenna ID	Azimuth
Airport Control Tower	GOV-ACT-BSU-004	09:25:43.0S, 160:02:56.1E	BSU-ACT-001	172.16.0.18	Cambium PMP 36320	WS-C3560CG-8PC-S	225

WIMAX INSTALL ---- See Appendix J / Section 18.1
MICROWAVE ALIGNMENT ---- See Appendix K / Section 19.1

7 Appendix A- How to use this Manual

This section explains the different types of installation you will be required to perform across different sites. The three types of installation sites are in the previous chapters six, seven and eight, containing Installation notes regarding each different site at the bottom of the page as per figure1 below.

INSTALLATION NOTES

MAST INSTALLATION ---See Appendix B / Category D –30ft Mast

CABLING ---See Appendix C / Section 11.1

NETWORKING (RACK) Internal---See Appendix E /Category C /Section 13.5

BUILDING ENTRY POINTS---See Appendix F / Section 14.1

LIGHTNING PROTECTION UNIT ---- See Appendix G / Section 15.1

ANTENNA ASSEMBLY ---- See Appendix I / Section 17.1

Figure 1

Refer to these Installation notes and select the instructions that pertain to your site as determined by category and section. Photocopy the pages, and then take them on site as a reference for correct installation.

Pay attention to visual notes as illustrated below as they there to aid you for a successful install.

 **NOTE** Check you have all the tools required prior to site install.

REFER TO 20 APPENDIX: L - TOOLS

The Appendix List is numbered and named according to their subject matter .In *Appendix B: Mast Installation* we have different Mast installation types listed via category as follows:

- Category: A - Hockey Stick
- Category: B – 10 foot Stay Mast
- Category: C - 20 foot Telescopic Mast
- Category: D - 30 foot Telescopic Mast
- Category: E - 40 foot Telescopic Mast

The next chapter is *Appendix C – Cable Installation*.

This is divided into three Sections providing detailed instructions on the cabling procedures as follows:

- 11.1 Cable Termination
- 11.2 Cable Termination PTP
- 11.3 CSM Ethernet Cable Connection

Appendix: D – Rack Unit (RU) Design

There are four types of locations requiring installation, as listed:

- Base Station Locations
- High Throughput Locations
- CPE Locations
- Cabled Locations

Following on we have *Appendix: E – Networking hardware – Layouts Rack Units (RU)*

Visual presentations depict the internal rack (RU) layout of the network hardware. This section is divided up based on the Locations as outlined in Appendix D. The contractor is to check that all network components have been sourced and ready for installation on site.

Appendix: F – Build Entry Point Installation

This section details the procedure for creating and sealing an opening suitable for passing cabling from the exterior to the interior of a building. The instructions show an entry point can be made either on a roof or through a wall.

Appendix G – Lightning Protection Unit (LPU) Installation

Illustrations depict step by step how to install the Lightning Protection Unit (LPU) to a mast.

Appendix H – Earthing

Earthing requirements notes for the PTP600/PMM 320 outdoor units. Also included are a set of General electrical notes and earthing notes for Communication Towers.

Appendix I –Antenna Assembly Install to Mast

Step by step procedures on how to install PTP600/PMP320 and CSM antennas.

Appendix J – WiMAX Install

Listed under this heading is the build of materials (BOM) needed for the wireless mesh installation. They are listed accordingly via installation categories ranging from masts to cabling.

Appendix K – Microwave Antenna PTP 600 Alignment

All antennas need to be aligning for optimum performance in link across to other antennas in the network. A step by step procedure details how to accomplish this task.

Appendix L – Tools

The tools used on site. Some tools are provided by CBO telecommunications and are listed in a checklist alongside illustrations of the tools. Equipment not provided are listed as Not Supplied

Appendix N – Glossary

The Glossary section provides a definition of any terms, abbreviations or acronyms used within this manual.

8 Appendix B - Mast Installation

This section details the construction of the mast to be used on site. Select the mast that pertains to your site as determined by category. Once selected photocopy the pages which are to be taken on site as a reference for correct installation:

REFER TO APPENDIX: L - 20.1 Tool Bag

 **NOTE** Do not go on site unless you have all the tools required.

8.1 Appendix B - Category: A - Hockey Stick

10.1 Category: A - Hockey Stick Installation Instructions – Sheet 1/1

STEP 1: Materials



Figure1

1. Silicone gun
2. Magnetic level
3. Cordless drill
4. Tube Clear Silicone
5. Drill set
6. Pen marker
7. Grinder (Hack saw if required)
8. 17mm Socket & wrench
9. Tape Measure
10. Hockey Stick Mast
11. 2 x Coach Bolts– 45mm 1/2" to 3/8" WW
12. Hammer drill (only needed if roof is concrete) 2 x Dynabolts
13. 2" x 3/8" (50mm x 9.5mm) (only needed if roof is concrete)

STEP 2 Source location

Place hockey stick in location for installation.

Make sure mast is level.

Mark out two holes on the fascia board or wall from the mounting plate. This is for the screws to be located.



Figure 2

STEP 3 Installation of mast

Drill two holes where marked. Depth should be approx. 40mm.

Place silicone around the outside of the two holes.

Using coach bolts screw in the hockey stick into place. Make sure the hockey stick is level.

Note: Mounting plate has four holes. If the first two bolts aren't strong enough, drill holes and screw in two more bolts for extra strength.

If mounting the hockey stick to concrete or a similar material, will need to use dynabolts for the mounting.

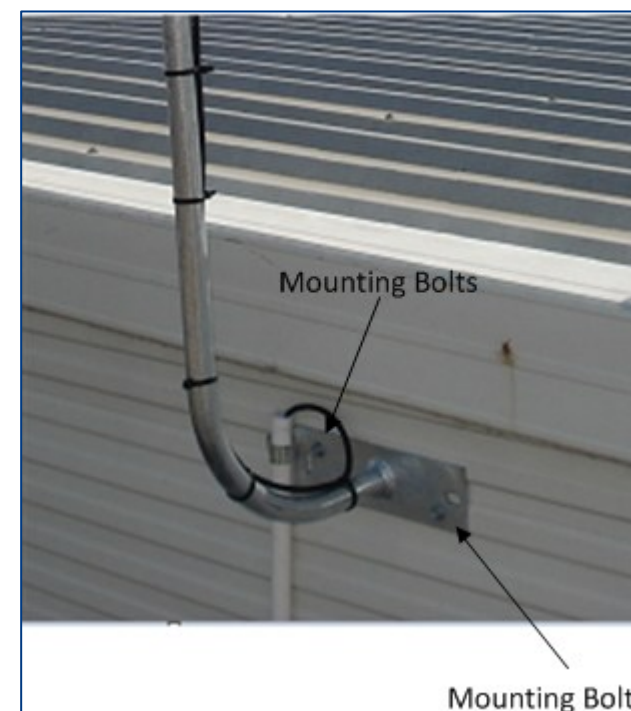


Figure 3

STEP 4 Installation of mast

Mark out length of hockey stick required and trim off excess with cordless grinder. Mast is ready for antenna mounting.

Place silicone over the bolts head for protection.



Figure 4

If you have any queries or concerns please call **1300 CBO NET** (1300 226 638).

Appendix B - Category: D - 30 foot Mast Installation Instructions – Sheet 2/3



Figure 8

Note: Silicone is placed around the hole and on screw heads to stop water leaking into the roof.

If roof type is concrete then two dynabolts are required for the roof mount. Each anchor points require one dynabolt.

STEP 8 Install first set of Anchor points

From the roof mount measure out 3m in all four directions shown in Figure 9& 11 with measuring tape. Remove roof screws and place silicone around holes. Screw in anchor points and seal with silicone over screw heads. Anchor points should be facing the roof mount.

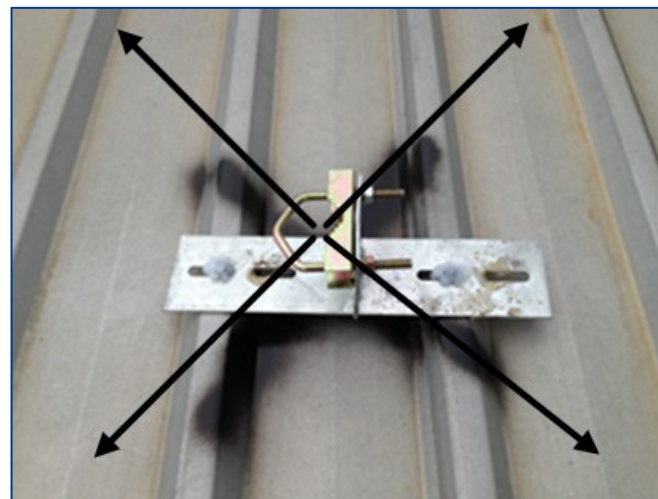


Figure 9

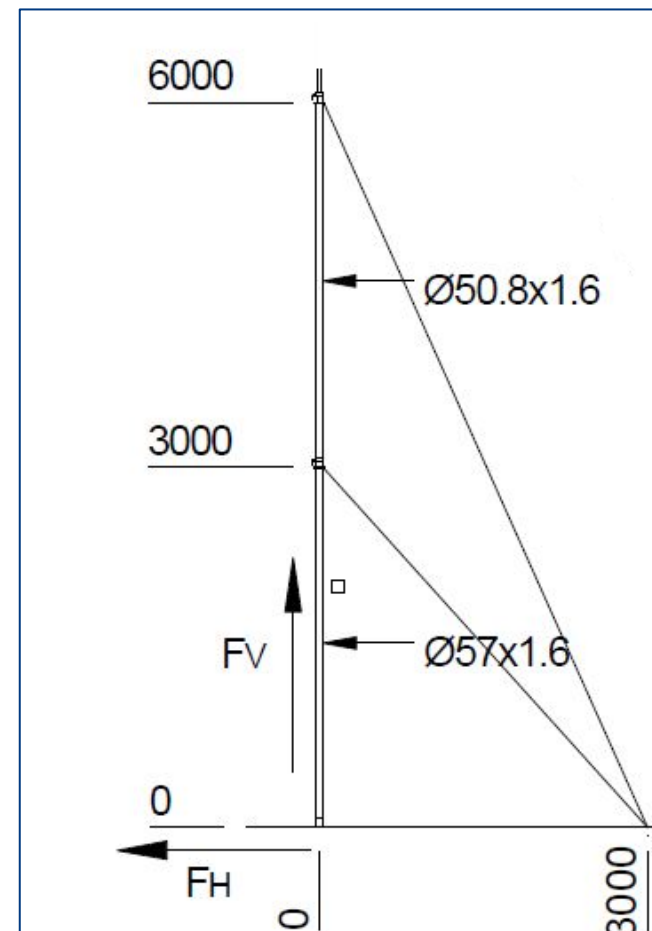


Figure 10

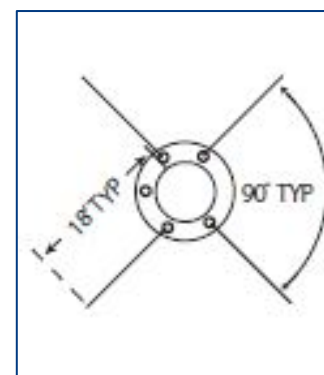


Figure 11

STEP 9: Install second set of Anchor points

From the roof mount measure out 6m in all four directions shown in Figure 9& 11 with measuring tape. Remove roof screws and place silicone around holes. Screw in anchor points and seal with silicone over screw heads. Anchor points should be facing the roof mount.

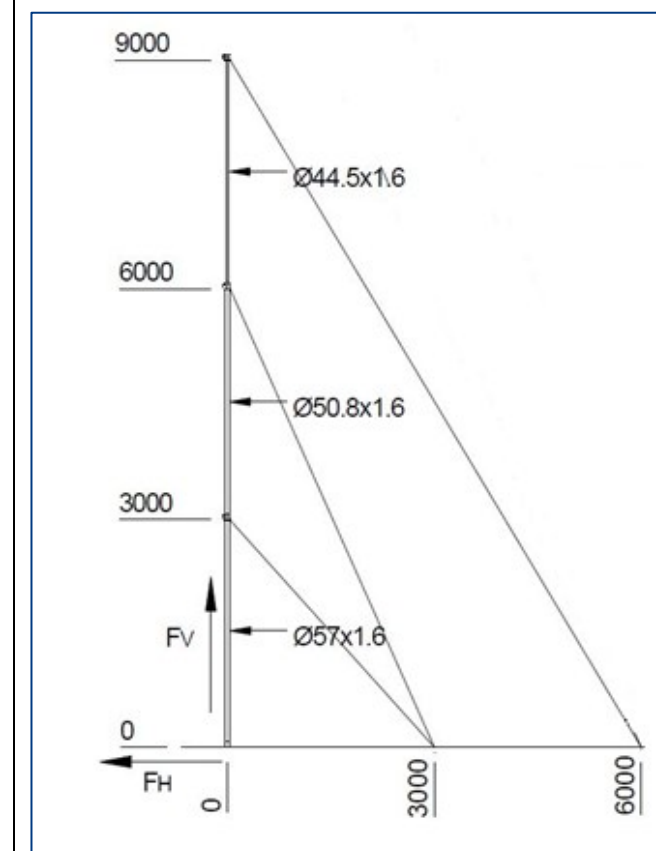


Figure 12

STEP 10: Install Mast

Place mast into the roof mount. 2nd person tighten up the two nuts on the roof mount. This will hold the pole in place at the base only.

STEP 11: Fasten up First set of guy ropes

Place turnbuckle with the hook down in the anchor point. Place thimble in the eye of the turnbuckle. Place guide rope through the eye of the turnbuckle and pull through excess wire rope.

Fasten up the wire grips on the wire rope as per Step 4

Align the mast vertically by tightening the Guy Lines.

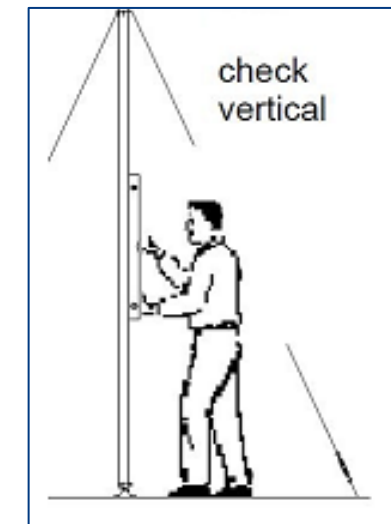


Figure 13

STEP 12: Raise second section of mast

Untie all of the top four wire ropes from the top section of mast. Push up one section and insert the retaining pins. Lock off the screw on the lock ring.

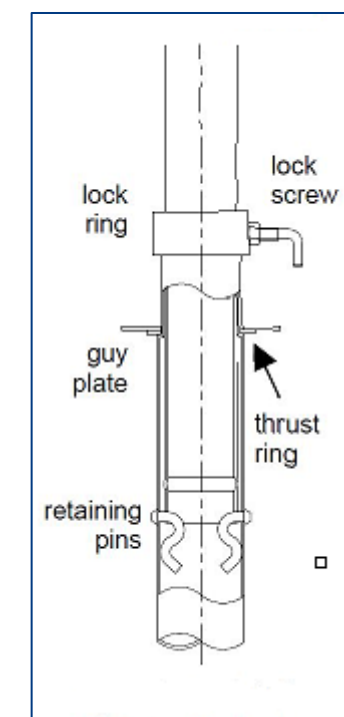


Figure 14

8.2 Appendix B - Category: E - 40 foot Telescopic mast

Category: E - 40 foot Telescopic mast Installation Instructions – Sheet 1/3

STEP 1: Materials



Figure1

1. Silicone gun
2. Magnetic level
3. Cordless drill
4. Tube Clear Silicone
5. Hex Drill set – 8mm & 10mm
6. 14mm & 13mm Box/Open end Spanner
7. Spray Marker
8. 1 x Roof Mount
9. 1 x 20ft Telescopic Mast
10. 4 x Guy Anchor Points
11. 16 x Thimbles
12. 8 x Turnbuckle Hook & Eye
13. 48 x wire grips
14. Rope Grip Wire
15. Metric tape Measure
16. Hammer drill (only needed if roof is concrete)
17. 10 x Dynabolts 2" x 3/8" (50mm x 9.5mm) (only needed if roof is concrete)

STEP 2 Source location

Mark out the area where the mast is to be located. This is where the Roof Mount will be placed. Refer to figure 2.

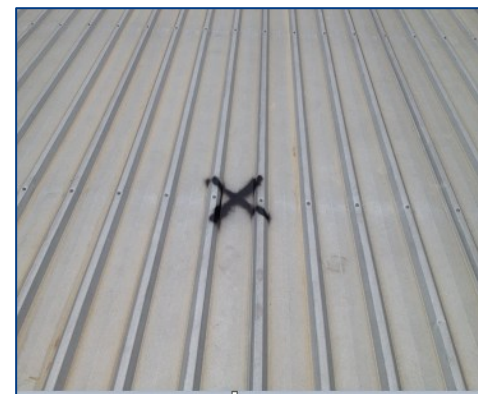


Figure 2

STEP 3 Prepare Collars on Mast – Fourth, Third, Second & First Section

Lay mast in a suitable area and extend mast out to full length. Place four thimbles on each collar. There are 4 collars in total.



Figure 3

STEP 4 Prepare Wire Rope – Fourth Section

Starting with the forth section place wire rope around thimble. Feed wire rope from underneath collar and allow approx. 200mm of cable. Attach wire grips; at least three wire grips should be used. Wire rope grips should be spaced at a distance of approximately six rope diameters with the saddles on the live part of the rope, and the U-bolt pressing on the less heavily loaded tail of the rope, see Figure 4 & 5 for reference.

Once completed, step out the wire rope till it reaches the bottom of the mast. Step out 7m from the bottom of the mast at 90°. Now cut the wire rope. Wind the wire rope up and tape to top of the fourth section of the mast. Continue this for the other three thimbles on the top collar.

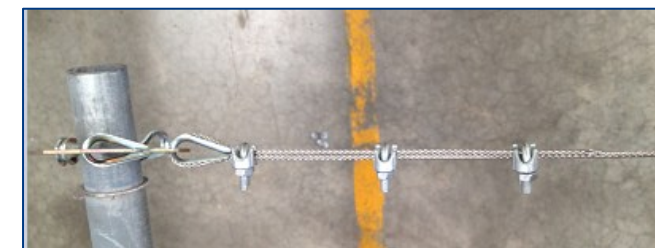


Figure 4

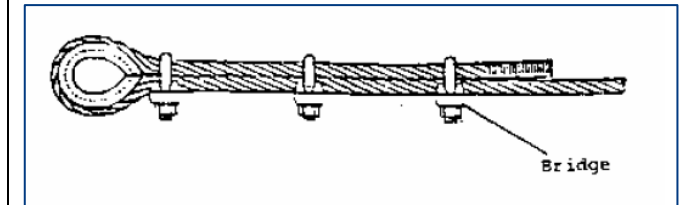


Figure 5

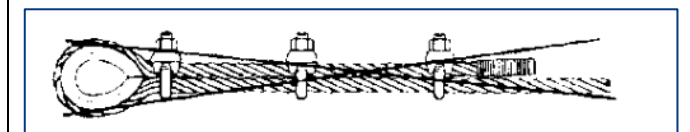


Figure 6

Note: Do not fit any or all of the wire grips with the bridge on the side opposite to the working part of the wire rope

STEP 5: Prepare Mast – Second Section

Repeat Step 4 for third section. Third second step out 7m.

STEP 6 Prepare Mast – Second Section

Repeat Step 4 for second section. Second section to step out to 4m.

STEP 7: Prepare Mast – First Section

Repeat Step 4 for first section. First section step out 4m.

STEP 8 Install Roof Mount

Remove two existing roof screws from the roof. Place silicone around the two holes. Place the roof mount over the holes and screw the two roof screws back into place. Place silicone over the screw heads for extra protection

Appendix B - Category: E -40 foot Mast Installation Instructions – Sheet 2/3



Figure 7

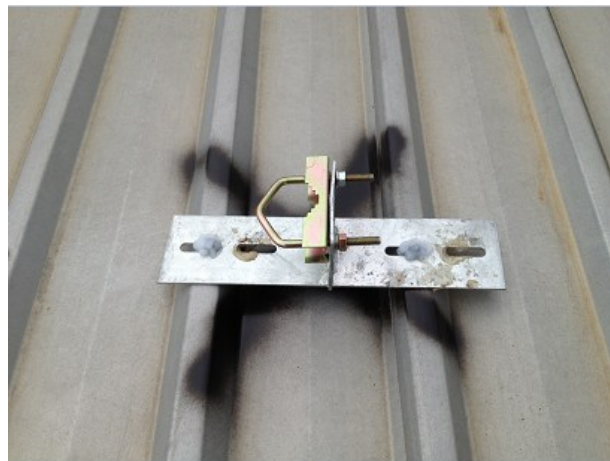


Figure 8

Note: Silicone is placed around the hole and on screw heads to stop water leaking into the roof.

If roof type is concrete then two dynabolts are required for the roof mount. Each anchor points require one dynabolts.

STEP 9 Install first set of Anchor points

From the roof mount measure out 3m in all four directions shown in Figure 9& 11 with measuring tape. Remove roof screws and place silicone around holes. Screw in anchor points and seal with silicone over screw heads. Anchor points should be facing the roof mount.

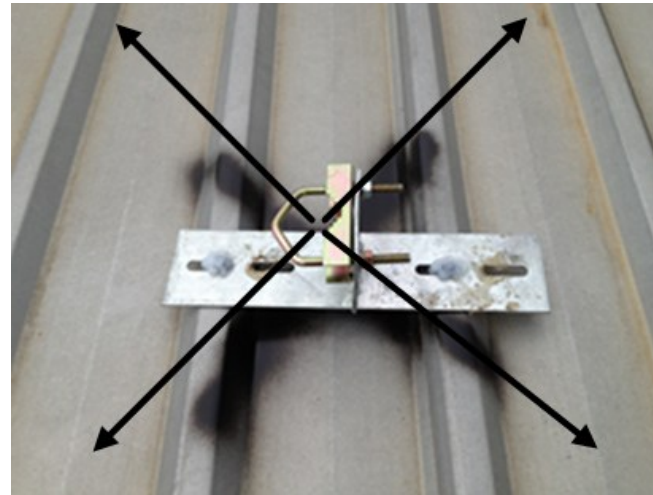


Figure 9

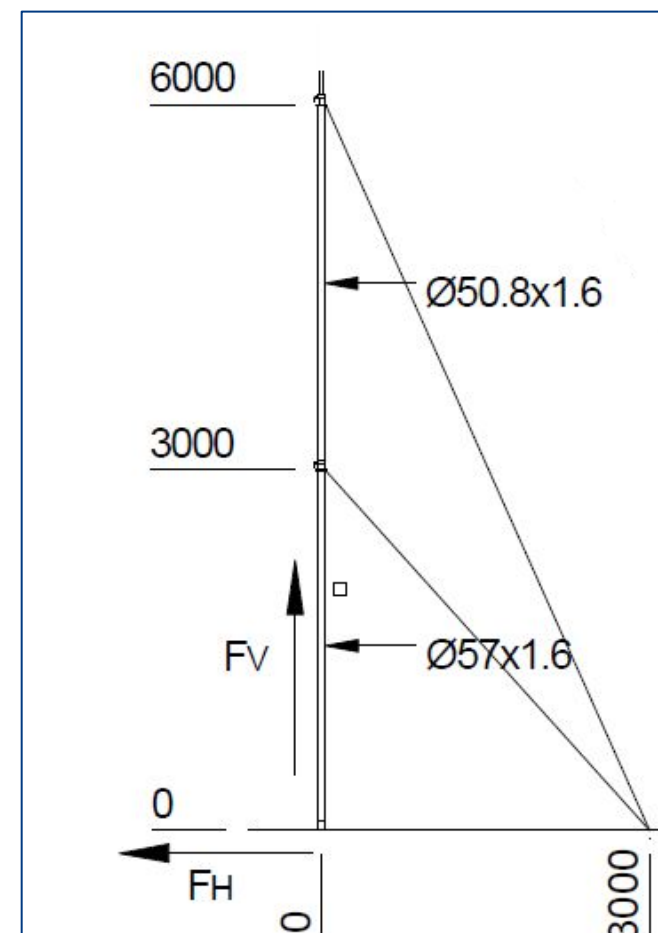


Figure 10

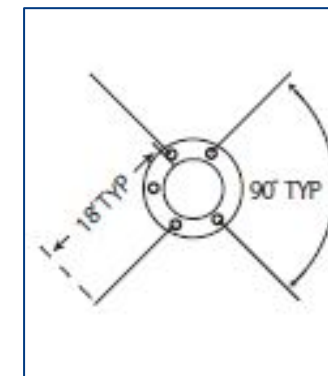


Figure 11

STEP 10: Install second set of Anchor points

From the roof mount measure out 6m in all four directions shown in Figure 9& 11 with measuring tape. Remove roof screws and place silicone around holes. Screw in anchor points and seal with silicone over screw heads. Anchor points should be facing the roof mount.

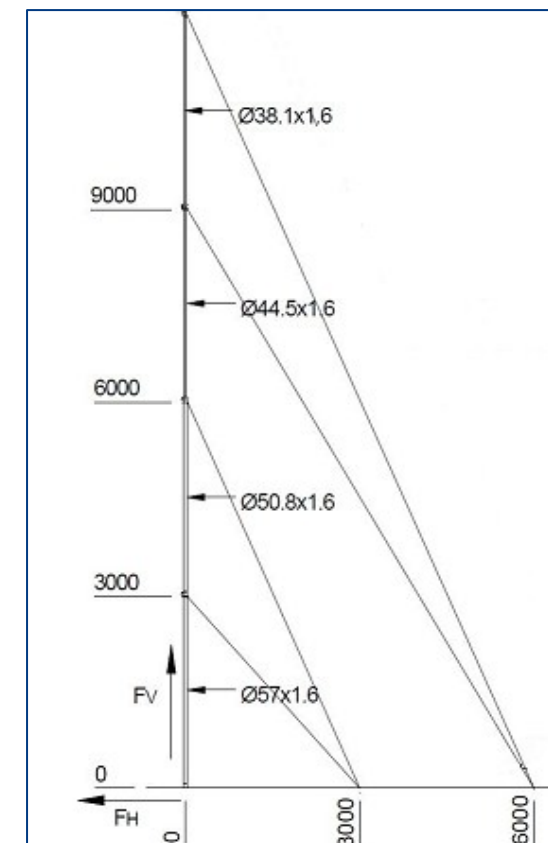


Figure 12

STEP 11: Install Mast

Place mast into the roof mount. 2nd person tighten up the two nuts on the roof mount. This will hold the pole in place at the base only

STEP 12: Fasten up First set of guy ropes

Place turnbuckle with the hook down in the anchor point. Place thimble in the eye of the turnbuckle. Place guide rope through the eye of the turnbuckle and pull through excess wire rope.

Fasten up the wire grips on the wire rope as per Step 4

Align the mast vertically by tightening the Guy Lines.

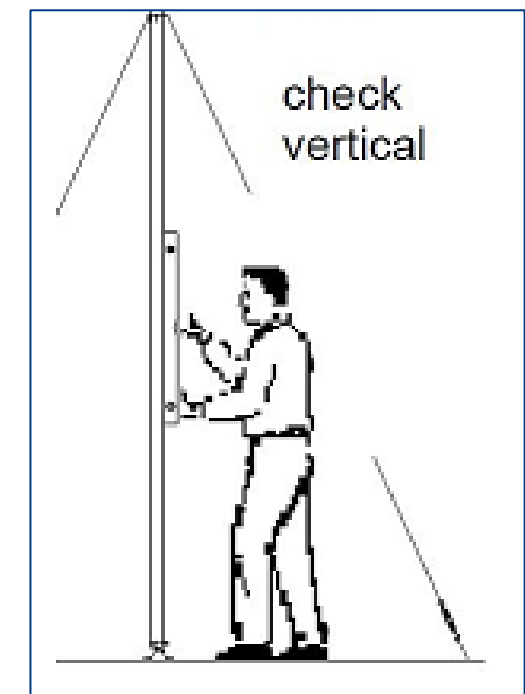


Figure 13

STEP 11: Raise second section of mast

Untie all of the top four wire ropes from the second section of mast. Push up one section and insert the retaining pins. Lock off the screw on the lock ring.

Appendix B - Category: E -40 foot Mast Installation Instructions – Sheet 3/3

STEP 12: Fasten up First set of guy ropes

Place turnbuckle with the hook down in the anchor point. Place thimble in the eye of the turnbuckle. Place guide rope through the eye of the turnbuckle and pull through excess wire rope.

Fasten up the wire grips on the wire rope as per Step 4
Align the mast vertically by tightening the Guy Lines.

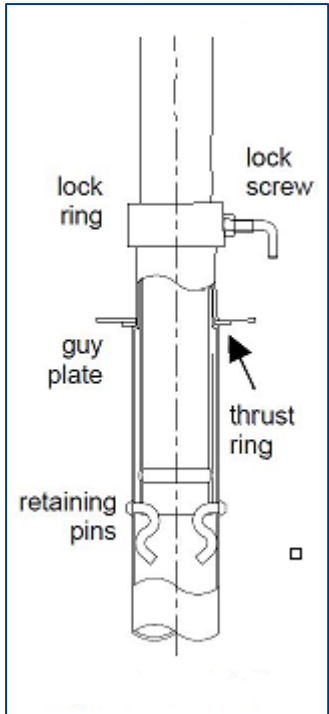


Figure 14

STEP 14: Fasten second set of guy ropes

Repeat Step 9 for the second section of the mast.

STEP 15: Raise third section of mast

Repeat Step 10 for the third section of the mast.

Note: EWP or platform ladder is required.

STEP 16: Fasten third set of guy ropes

Repeat Step 9 for the fourth section of the mast.

STEP 17: Raise fourth section of mast

Repeat Step 10 for the fourth section of the mast.

STEP 18: Fasten fourth set of guy ropes

Repeat Step 9 for the fourth section of the mast.

Note: EWP or platform ladder is required.

STEP 19: Fasten third set of guy ropes

Stand at base of mast and look up the mast. Align the mast vertically by tightening the respected Guy Lines.

For more detail align step away and use a theodolite machine which will guarantee the mast to be straight.

Note: With the theodolite make sure you take an alignment from one position then another from 90° from the original position.

STEP 20: Helpful Hints

Never attempt to rise up, or lower, a telescoping mast during a thunderstorm in your vicinity. Even if lightning doesn't strike the mast directly, a nearby strike can still cause electrocution and possibly death.

If you have any queries or concerns please call **1300 CBO NET**
(1300 226 638).

9 Appendix C - Cabling Installation

This section details the work procedures required for installing cabling to be used on site. Select the work instruction that pertains to your site. Once selected photocopy the pages which are to be taken on site as a reference for correct installation:

Ensure you have all tools needed for the site.

REFER TO APPENDIX: L/ 20.1 Tool Bag

 **NOTE** Do not go on site unless you have all the tools required.

General electrical notes

To comply with the following.

1. All of the electrical installation shall comply with the requirements of AS3000-wiring rules. The relevant supply authority
2. All cabling is to be installed in such a manner that is readily drawn or replaced after construction,
3. All penetrations are to be made good to pre-existing conditions.
4. All wiring is to be concealed except in plant rooms, switch rooms and electrical riser enclosures.
5. Cable routes are to be determined and verified by the contractor on site.
6. The contractor is to liaise with the supply authority to coordinate and confirm the main installation works.
7. The cable from the power company to the equipment shelter or fit-out room shall be certified correct before connection and energisation in accordance with AS3000 and regulatory requirements.
8. Underground services: the location of existing services is a schematic representation of information contained in drawings issued by the relevant authorities. Additional undocumented services may be present on site; the installer shall identify and confirm the location of all relevant underground services prior to commencement of any works by contacting the relevant resource.
9. All equipment and cabling is to be labelled.
10. All material and equipment used shall comply with the latest Australian standards for that material or equipment.

9.1 Cable Termination

Appendix C – Category A - Cable Termination TIA/EIA – 568 Standard (Sheet 1/1))

STEP 1: Materials

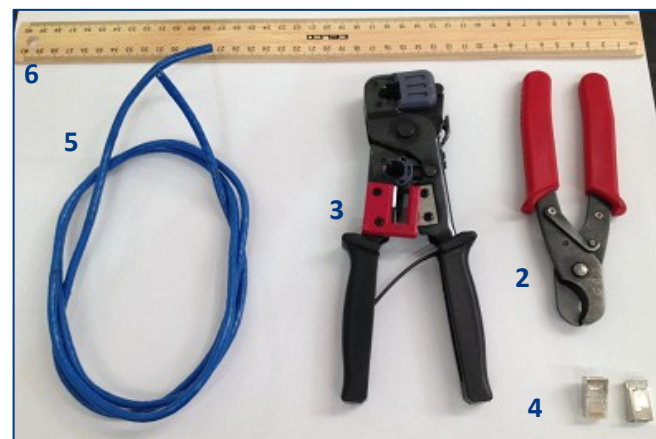


Figure 1

Materials List

1. Wire Strippers (Not pictured)
2. Wire Cutters (Side Cutters will work)
3. RJ45 Crimping Tool
4. 2x RJ45 Shielded Modular Data Plug (Ends)
5. Bulk CAT5e Network Cable
6. Ruler
7. 2 Wire Boots (Optional, not pictured)
8. Gland from Point-to-Point 600 kit (Not pictured)

STEP 2 Cable Preparation

Using a **Crimping Tool**, trim the end of the cable you're terminating, to ensure that the ends of the conducting wires are even.



Figure 2

STEP 3 Measure/Trim Cable

Being careful not to damage the inner conducting wires, strip off approximately 1 inch of the cable's jacket, using a **modular crimping tool** or a **UTP cable stripper**.



Figure 3

STEP 4 Prepare Wires

Separate the 4 twisted wire pairs from each other, and then unwind each pair, so that you end up with 8 individual wires. Flatten the wires out as much as possible, since they'll need to be very straight for proper insertion into the connector.

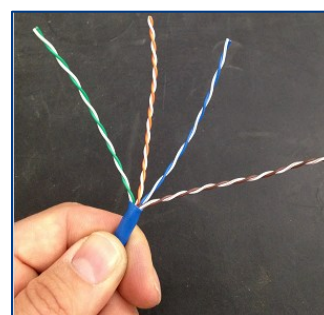


Figure 4

STEP 5

Holding the cable with the wire ends facing away from you. Moving from left to right, arrange the wires in a flat, side-by-side ribbon formation, placing them in the following order: **white/green, solid green, white/orange, solid blue, white/blue, solid orange, white/brown & solid brown.**

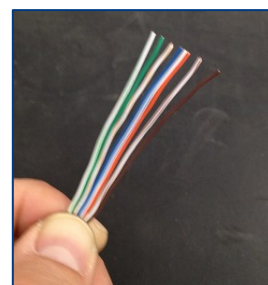


Figure 5

STEP 6 Prepare Connector

Holding the RJ45 connector so that its pins are facing away from you and the plug-clip side is facing down; carefully insert the flattened, arranged wires into the connector, pushing through until the wire ends touch the end of the connector. For strength of connection, also push as much of the cable jacket as possible into the connector.

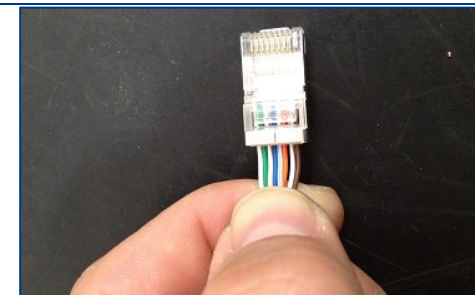


Figure 6

STEP 7 Prepare for Crimping

Check to make sure that the wire ends touch the end of the connector. Also check the wires are in the correct order; if not, remove them from the connector, rearrange into proper formation, and re-insert. Remember, once the connector is crimped onto the cable, it's permanent. If you realize that a mistake has been made in wire order after termination, you'll have to cut the connector off and start all over again!

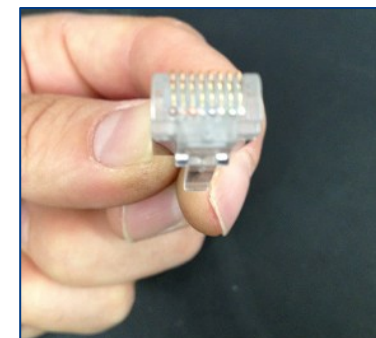


Figure 7

STEP 8 Check Termination

Insert the prepared connector/cable assembly into the 8 port RJ45 slot in your crimping tool. Firmly squeeze the crimper's handles together until you can't go any further. Release the handles and repeat this step to ensure a proper crimp.

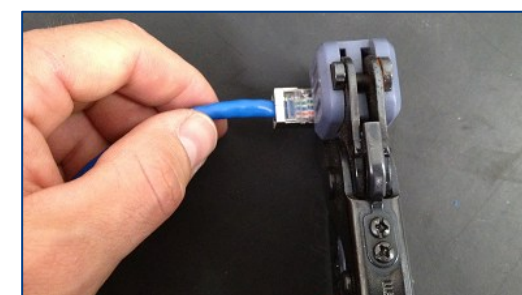


Figure 8

STEP 9 Check Termination

After the first termination is complete, repeat process on the opposite end of your cable

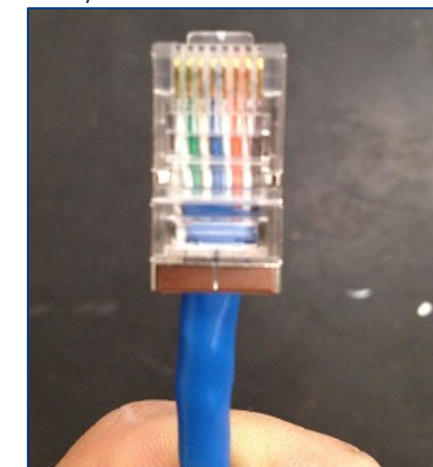


Figure 9

STEP 10 Verify and Test Cable

Once completed both ends verify cable pin outs and then test with a LAN Tester.

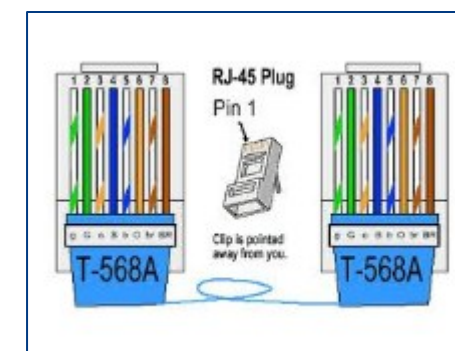


Figure 10

If you have any queries or concerns please call

1300 CBO NET (1300 226 638).

10 Appendix D – Rack Unit Design

10.1 Base Station Locations

Each base station location has been allocated a RU floor-based lockable data cabinet. This rack is essential to house UPS' and extra UPS batteries. A 24 port patch lead will have all cables terminated, a high end L2/L3 Cisco 8 port switch as well as the CMM to provide power to the radio equipment. The UPS design will allow up times no less than 2 hours and we will exceed these expectations. Note Anthony Saru and Disaster Recovery Tower have existing backup systems and available space. . This equipment has been moved to allow for higher uptimes at other sites.

10.2 High Throughput Locations

Each base station location has been allocated a RU wall mounted lockable data cabinet. The wall cabinet is essential to house UPS' and provide uptime over 40 minutes and above. A 24 port patch lead will have all cables terminated, a high end L2/L3 Cisco 8 port switch as well as the CMM to provide power to the radio equipment. The UPS design will allow up times no less than 2 hours and we will exceed these expectations. Note Anthony Saru and Disaster Recovery Tower have existing backup systems and available space. This equipment has been moved to allow for higher uptimes at other sites.

10.3 CPE Locations

Each base station location has been allocated a RU wall mounted lockable data cabinet. The wall cabinet is essential to house the UPS, which provides uptime of forty minutes and above. A 24 port patch panel will have all cables terminated, a high end L2/L3 Cisco 8 port switch as well as the CMM to provide power to the radio equipment. The UPS design will allow up times no less than 2 hours and we will exceed these expectations. Note: the Anthony Saru Building and the Disaster Recovery Tower have existing backup systems and available space. This equipment has been moved to allow for higher uptimes at other sites.

10.4 Cabled Locations

These locations are situated in the same buildings already serviced by another department. A single Cat5 cable terminated in the existing data cabinet labelled MAN will be terminated to this already existing service. A switch has been allowed for all existing CPE sites to give the ability of multiple offices/departments logging into the network for current and future use.

10.5 General electrical notes

To comply with the following:

1. All of the electrical installation shall comply with the requirements of AS3000-wiring rules.
The relevant supply authority
2. All cabling is to be installed in such a manner that is readily drawn or replaced after construction,

3. All penetrations are to be made good to pre-existing conditions.
4. All wiring is to be concealed except in plant rooms, switch rooms and electrical riser enclosures.
5. Cable routes are to be determined and verified by the contractor on site.
6. The contractor is to liaise with the supply authority to coordinate and confirm the main installation works.
7. The cable from the power company to the equipment shelter or fit-out room shall be certified correct before connection and energisation in accordance with AS3000 and regulatory requirements.
8. Underground services: the location of existing services is a schematic representation of information contained in drawings issued by the relevant authorities. Additional undocumented services may be present on site; the contractor shall identify and confirm the location of all relevant underground services prior to commencement of any works by contacting the relevant resource.
9. All equipment and cabling is to be labelled.
10. All material and equipment used shall comply with the latest Australian standards for that material or equipment.

11 Appendix E –Networking Hardware - Layouts Rack Units (RU)

This section illustrates the installed layout of networking hardware for the Communications Rack Units (RU) on site. Select the instructions that pertain to your site as determined by category. Photocopy the pages, and then take them on site as a reference for correct installation:

11.1 Category A Site: BSU (Base Site Unit) Rack Layout (Internal)

11.1.1 BSU Location: SATSOL-TX-BSU-001 - Slave

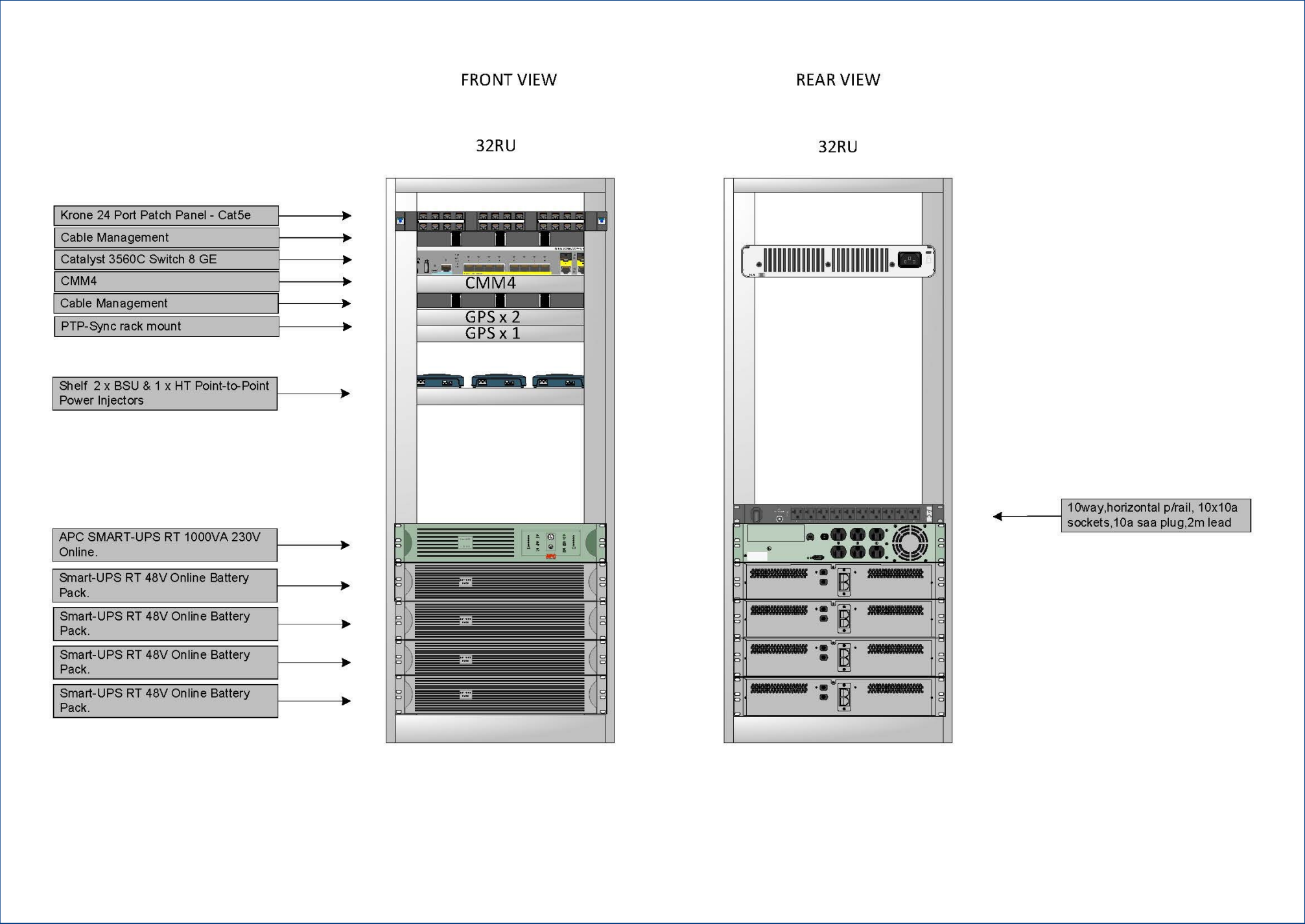


Figure 1 – Location of Networking Hardware in-SI-NDMO-BSU-003

11.1.2 BSU Location: GOV-POL-BSU-002 - Slave

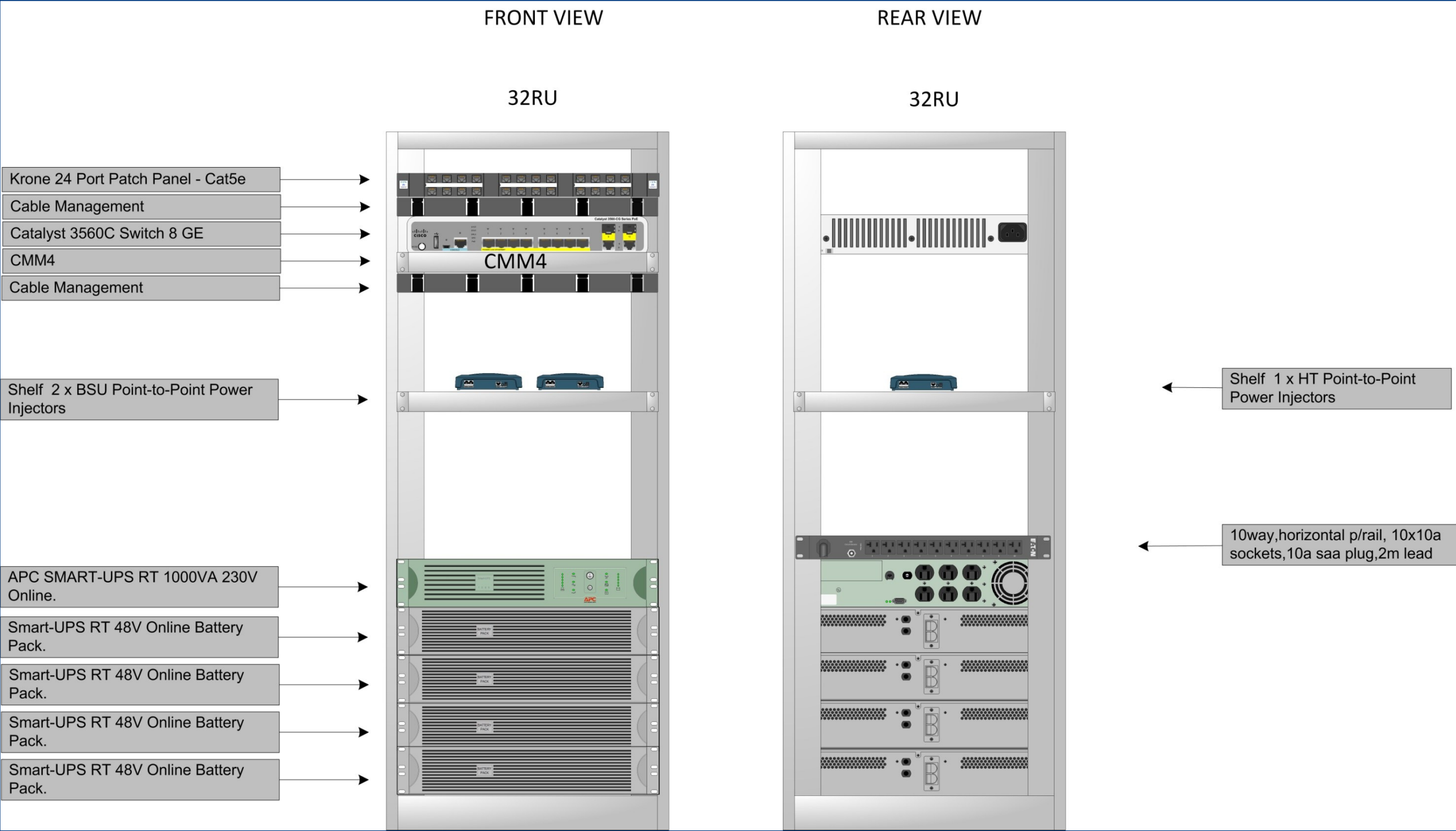


Figure 2 – Location of Networking Hardware in-SI-NDMO-BSU-003

12 Appendix F – Building Entry Point Installation

This section provides necessary information about creating an entry point in a surface for cabling. Has detailed procedure for creating and sealing an opening for cable entry. Suitable for passing cabling from the exterior to the interior of a building. An entry point can be found either on a roof or through a wall.

The step by step work instructions on the following pages are to be followed to meet current industry standards. Photocopy the pages, and then take them on site as a reference.

Ensure you have all tools needed for the site. Refer to the illustrated Materials list –Step 1 on the work instruction.

REFER TO APPENDIX: L - 20.1 Tool Bag



Do not go on site unless you have all the tools required.

12.1 Appendix F - Category A Site: - Building Entry Point Instructions Sheet 1/1

STEP 1: Materials



Figure 1

1. Tube of clear silicone
2. Silicone gun
3. Cordless drill
4. Tinsnips
5. 1 x Mini dektite
6. Washered self-drilling Screws
7. Hex drill piece (for screws)
8. Hammer drill (only needed if roof is concrete) (Not pictured)
9. 10 x Dynabolts 2" x 3/8" (50mm x 9.5mm) (only needed if roof is concrete)

Helpful Hints

When cutting, do not use a knife.
Sharp tin snips will provide the smoothest finish.
Always cut where sizes are indicated.
Hand form aluminium edge before fastening.
Fasten from vertical centre to outside

STEP 2: Prepare Dektite

Simply cut Mini Dektite cone to required size for 'snug' fit



Figure 2

STEP 3: Installing Dektite

Push or pull cable or conduit through cone.



Figure 3

STEP 4: Sealing Dektite

Apply neutral-cure clear Silicone to underside of flashing (embossed ribbed section). If necessary press flashing into contours of surface configuration



Figure 4

STEP 5 Fastening Dektite

Fasten flashing to wall or roof surface with washered self-drilling screws.

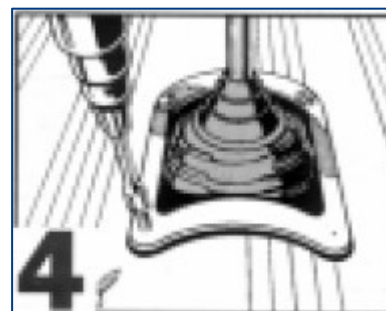


Figure 6

STEP 6 Install Stays to Mast end

Place neutral-cure clear silicone around the edges of the dektite and also the around the entry point of the cable.

If you have any queries or concerns please call
1300 CBO NET (1300 226 638).

13 Appendix G – Lightning Protection Unit (LPU) Installation

This section provides necessary information about installing an entry point in a surface for cabling. The work instructions on the following page are a step by step process that needs to be followed to meet current industry standards. Photocopy the pages as required, and then take them on site as a reference.

Ensure you have all tools needed for the site.

REFER TO APPENDIX: L - 20.1 Tool Bag



Do not go on site unless you have all the tools required.

13.1 Appendix G – PTP 600 LPU Installation

Appendix G – Section 15.1 / PTP 600 - Lightning Protection Unit Install Instructions (Sheet 1/1)

STEP 1: Materials

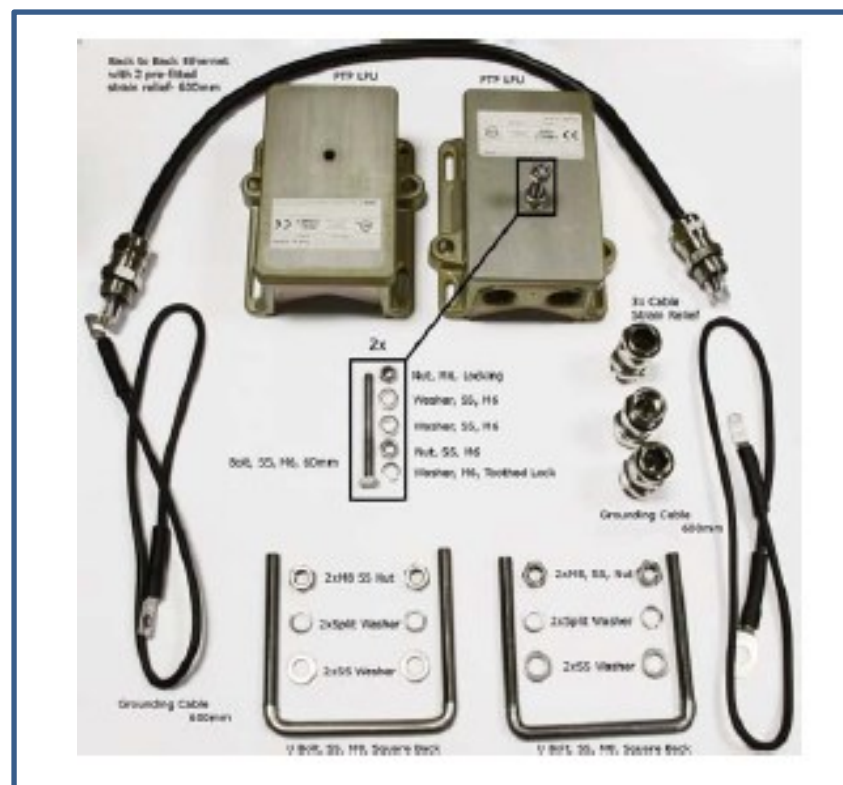


Figure 1 – PTP LPU Kit

Materials List

- PTP LPU kit

STEP 2: Assemble of LPU

Check that all parts of the PTP LPU kit are in the box.

STEP 3: Preparation of the LPU

Decide how the LPU will be mounted; refer to Figures 2 & 3 for installation examples

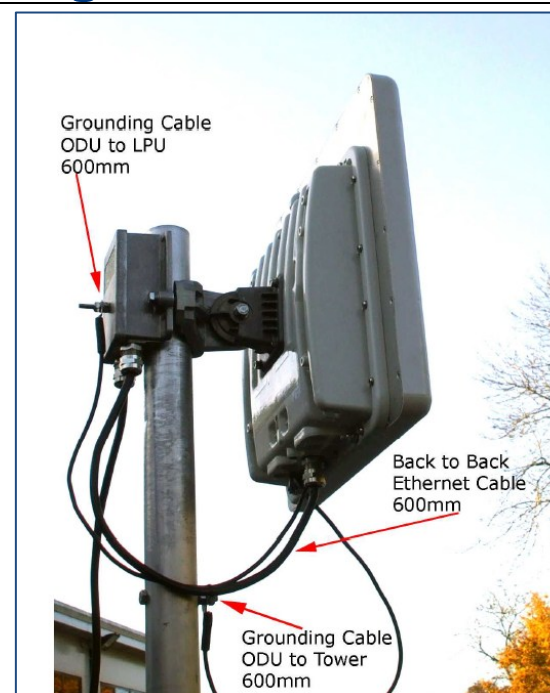


Figure 2 – Installation Option 1 – Back to Back with ODU



Figure 3 – Installation Option 2 – LPU U-Bolt Mounting

STEP 4: Mounting the LPU

Mount the LPU vertically with cable glands facing downwards.
Use the supplied Ethernet cable to connect the LPU to the ODU.
Use the cable glands provided and the recommend Ethernet cable to complete the installation; refer to Figures 4 for correct cable preparation.

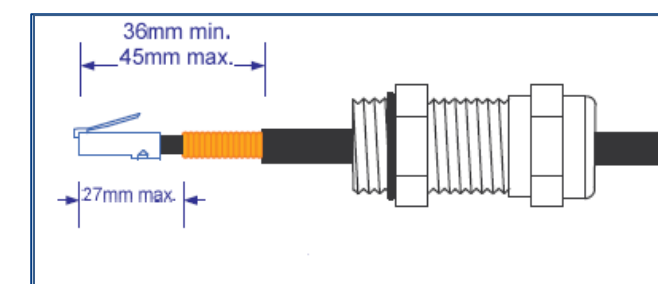


Figure 4

STEP 5: Connecting the Cable

Plug in the RJ45 into the LPU and screw the gland body into the LPU and tighten, before getting the back shell of the gland.

Do not over tighten the gland back shell; refer to Figure 5.



Figure 5

Place neutral-cure clear silicone around the edges of the dektite and also the around the entry point of the cable.

If you have any queries or concerns please call **1300 CBO NET** (1300 226 638).

13.2 Appendix G – Surge Suppressor Installation for PMP & CSM

Appendix G – Section 15.3 / Surge Suppressor Installation Instructions (Sheet 1/1)

STEP 1: Materials



Figure 1 Surge Suppressor Unit

Materials List

- Surge Suppressor Unit

STEP 2: Attach LPU

Attach the surge suppressor in location inside communication hut.

STEP 3: Preparation of the Surge Suppressor

Remove the surge suppressor front cover to access the Ethernet RJ-45 jacks.

STEP 4: Connecting Cables

Connect the Ethernet cables from the PMP/CSM to the Ethernet port (J1) on the surge suppressor.

Connect the Ethernet cable from the CMM to the Ethernet port (J2) on the surge suppressor.

All cables must exit through the allocated slots.

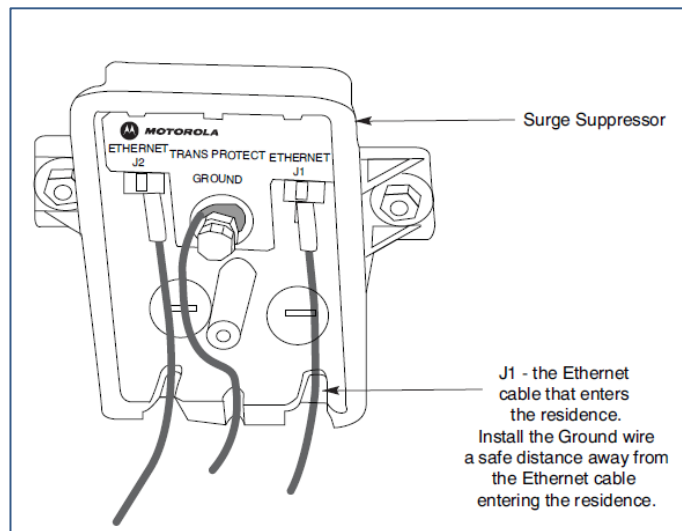


Figure 2

STEP 5: Grounding the Surge Suppressor

Ground the surge suppressor to the Earth ground.

If you have any queries or concerns please call
1300 CBO NET (1300 226 638)

14 Appendix H – Earthing

14.1 General electrical notes

To comply with the following:

1. All of the electrical installation shall comply with the requirements of AS3000-wiring rules. The relevant supply authority
2. All cabling is to be installed in such a manner that is readily drawn or replaced after construction,
3. All penetrations are to be made good to pre-existing conditions.
4. All wiring is to be concealed except in plant rooms, switch rooms and electrical riser enclosures.
5. Cable routes are to be determined and verified by the contractor on site.
6. The contractor is to liaise with the supply authority to coordinate and confirm the main installation works.
7. The cable from the power company to the equipment shelter or fit-out room shall be certified correct before connection and energisation in accordance with AS3000 and regulatory requirements.
8. Underground services: the location of existing services is a schematic representation of information contained in drawings issued by the relevant authorities. Additional undocumented services may be present on site; the installer shall identify and confirm the location of all relevant underground services prior to commencement of any works by contacting the relevant resource.
9. All equipment and cabling is to be labelled.
10. All material and equipment used shall comply with the latest Australian standards for that material or equipment.

14.2 PTP 600/PMP 320 Earthing requirements

1. Use shield Ethernet cable for all infrastructure cabling
2. Use drip loops at any point where cable changes direction
3. Extra cable (a splice loop) to be provided for future use at any termination.
4. Ensure before commencing work, structure is fully grounded (Protective Earth)
5. Use water tight tape, followed by electrical tape on all N-connectors

14.3 Tower Earthing notes

1. All equipment and antenna mounts are to be earthed.
2. All RF feeders shall be earthed at the top near the antenna using manufacturer provided earth kits and connected to masts.
3. All connections to earth bars shall be double lugged and snug tightened.
4. Mounting flange plate to be positioned 1.0m inside shelter to be using 35mm g/y
5. The earthing system shall be checked for continuity.

6. Ensure continuity of the entire length of the cable ladder. Where necessary bond each section together with 16mm² earthing cable.
 7. All earth bars inside cable ladders are to be bonded to the equipotent conductor.
 8. All connections to earth bars shall be double lugged and snug tightened.
 9. The earthing system shall be checked for continuity and its impedance shall be measured and a written test report provided.
 10. Earthing electrodes installed at a depths not less than 3m. Or otherwise in 75mm min. Drilled 4 x 100mm core holes to a depth of 4.0m. backfilled with ground enhancement compound (gypsum/bentonite mix.)
 11. Exterior earthing leads shall be as short as possible. All exterior earthing leads shall be connected vertically, tangential to the surface they are bonding and the cable run direct to the equipotential bonding conductor.
 12. All bends in earthing leads shall be installed with a minimum radius of 200mm.
 13. The quantities of earth stakes shown is a minimum. Install more earth stakes, as required to achieve the maximum resistance allowed, 5 ohm
 14. Generally earthing bars are required at the top and bottom of all vertical cable runs 3m or longer. Shorter vertical runs require only one earth bar. Horizontal runs require an earth bar at each end where the cable interfaces with the tower and shelter. Earth bars are also required where cable bends at more than 450 at not less than 3m apart and at the midpoint if runs are longer than 70m.
- .

15 Appendix I –Antenna Assembly Install to Mast

This section provides a guide to installing Microwave Units to masts on site. Select the instruction that pertains to your site. Once selected, photocopy the pages and take on site.

Refer to the illustrative guide as a reference for correct installation:

Ensure you have all tools needed for the site. **REFER TO APPENDIX: L - 20.1 Tool Bag**

 **NOTE** Do not go on site unless you have all the tools required.

15.1 Appendix I – Microwave Point to Point PTP 600 Installation to Mast

Appendix I – Microwave PTP600 Install to Mast Instructions

STEP: 1



Figure 1

Materials List

1. Outdoor Units (ODU)
2. Mounting Bracket Assembly
 - 'End Complete' kit contains one bracket
 - 'Link Complete' kit contains two brackets
3. Lightning Protection Unit (LPU) kits
4. 13mm Box/open spanners

STEP: 2 Mounting ODU Bracket

Attach the ODU bracket strap to the pole using M8 x 70 mm bolts, M8 flat washers and M8 coil washers.

The LPU must be vertical with cable glands facing downwards.

Tighten to ensure the assembly grips, but can be adjusted on the pole.



Figure 2

STEP 3

Use the integrated lifting eye to hoist the ODU up to the bracket

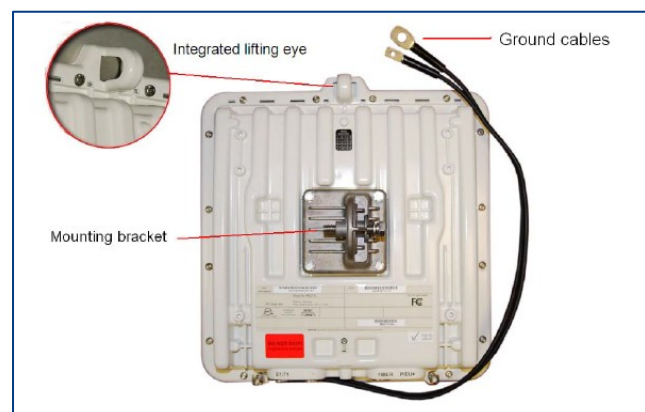


Figure 3

STEP: 3 Mounting the ODU

Offer the ODU (with pre-fitted mounting bracket) to the bracket strap and affix using the captive M8 bolt. Tighten to ensure the assembly grips, but can be adjusted on the pole using 13mm spanner.



Figure 4

STEP 5 Tightening the Brackets

Adjust the elevation and azimuth of the unit to achieve an approximate visual alignment. Tighten all three bracket bolts to the required torque setting of 14 Nm (11 lb. ft.).

If you have any queries or concerns please call **1300 CBO NET** (1300 226 638).

STEP 1: Materials



Figure 1

Materials List

1. 1x Motorola GPS unit
2. 1 x coil of GPS Cabling
3. M8 screws and washers (Not pictured)
4. U-bolts

Required Tools

- 13mm wrench and/or socket wrench for M8 nuts and bolts
- 10mm wrench and/or socket wrench for M6 nuts and bolts
- Needle-nose pliers
- Electrical tape for all GPS connectors
- Water-tight tape for GPS connections

STEP 2: Attach GPS Unit to bracket

Attach the GPS unit to the bracket base mount using 4 screws (provided)

STEP 3: Mounting GPS bracket to pole

Place the U-bolts (provided) around pole. Slide the GPS antenna bracket onto U-bolts. Slide the ring washers (provided) onto the U-bolts. Slide the lock washers (provided) onto the U-bolts. Use the nuts (provided) to securely fasten the bracket to the U-bolts; refer to Figure 2

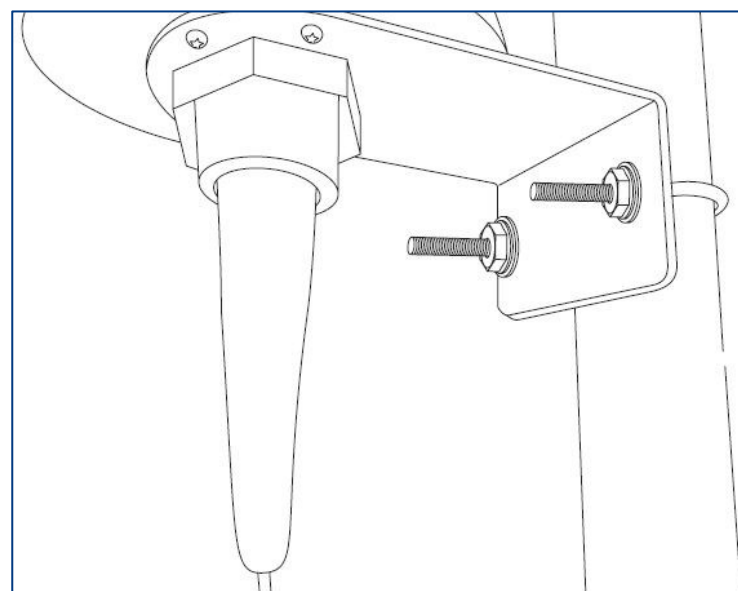


Figure 2

STEP 4 Connecting Coaxial Cable to GPS unit

Connect the GPS coax cable (provided) to the female N-connector on the GPS antenna; refer to Figure 3

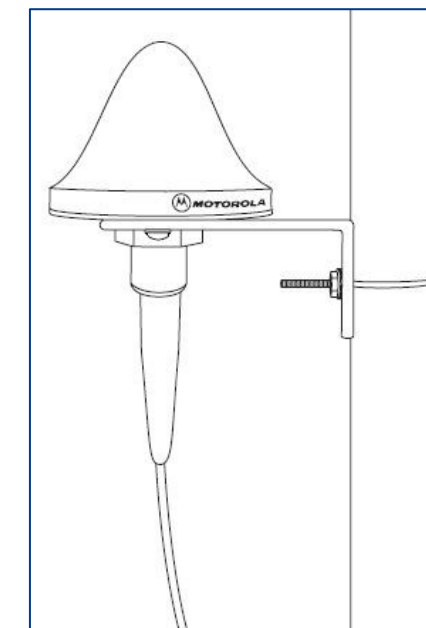


Figure 3

STEP 5 Sealing the connectors

Use waterproof tape and electrical tape on all N-type connectors on the antenna.

If you have any queries or concerns please call

1300 CBO NET
(1300 226 638).

16 Appendix J – WiMAX Install

16.1 Category A Hockey Stick Wi-Max CPE 370 mm Antenna - 1- 10 km Distance

Networking Hardware			
#	Equipment	Description	Quantity
1	3630SMC	PMP36320 Subscriber Module Connectorised	1
2	MA-WA36-DP21	3.3 - 3.8 GHz H/V Pol 22dBi 370x370 mm CPE Antenna with 2x N-Type, Female Connector Inc mount	1
3	WS-C3560CG-8PC-S	Catalyst 3560C Switch 8 GE, 2 x Dual Uplink, IP Base	1
4	C892FSP-K9	Cisco 892FSP Cisco 891 Gigabit Ethernet security router 10/100/1000 8 port + 2 WAN 10/100 ports	1

UPS Hardware			
#	Part Number	Description	Quantity
1	APAP9631	UPS Network Management Card 2 with Environmental Monitoring	1
2	SURT1000XLI	SMART-UPS RT 1000VA 230V Online.	1
3	SURTRK	Smart-UPS Online Rail Kit for 1000 and 2000VA models.	1

Install MISC.			
#	Part Number	Description	Quantity
1	11MM-HS48	Curved Fascia Mount 1.2m (40")	1

Install MISC.			
#	Part Number	Description	Quantity
2	11MM-HSS4FT	Hockey Stick 4FT (1.2m) Stay Bar (Pair)	1
3	11MM-UB8M/2	U-Bolts 2 x (4.25" x 5/16")	1
4	11MM-HC01	Hockey Stick (25-32mm) Stay Bar Collar	1
5	11MM-HDN	Heavy Duty Nest	3
	11MM-PC32	32mm PVC Pole End Cap	1
6		Assorted Screws (wood/roofing/metal)	
7	11MM-DB1050	Dynabolt 2" x 3/8" (50mm x 9.5mm)	6
8			
9	n/a	Black UVCat5e Outdoor FTP (m)	90
10	FPAS-CP20B	20mm Black FPAS flexible conduit (incl. 10 x fittings and 10 x locknuts)	1
11	11MM-CT275/100	Black Plastic Cable Ties (100 per pack)	1
12	n/a	RJ45 Shielded Solid Connector	20
13	11MM-SI01	Silicone Tube	1
14	CNDEKSM	Dektite 0 ~ 50mm	1
15	180HD20	20mm Hot Dip Galvanised Half Saddle	50
16	9020CMS20GY	Corrugated PVC Conduit. 20mm Medium Duty Sheathed	25
17		UV Insulation Tape	1
18		Self-amalgamating Tape	1
19	n/a	6mm2 earth wire (m)	10
20	n/a	6mm2 earth wire lugs	10

Rack			
#	Part Number	Description	Quantity
1	MFB0235904	CABINET SERIES 210 18RU 600MM DEEP	1
2	MFB0272901	CABINET WALLMOUNTING KIT - FOR SERIES 94 (Flat packed)	1
3	MFB0236610	DOOR SERIES 210 18RU ACRYLIC	1
4	MFB0301001	FIXED SHELF - 300D 19IN	1
5	MOLPID00227	PATCH PANEL CAT5E 24 PORT 1U + MGR	1
6	MFB9350040	POWERBOARD HORIZ 2RU 19IN 10 X 10A STD PLUG 2.5M L	1
7	6450 1 050-10	Patch Cord Minder	2
8	MFB0235904	CABINET SERIES 210 18RU 600MM DEEP	1

Cable			
#	Equipment I	Description	Quantity
1	6451 5 039-05	Cat5E RJ45-RJ45 Ethernet Patch Cable	2

16	n/a	RJ45 Shielded Solid Connector	20
17	11MM-SI01	Silicone Tube	1
18	CNDEKSM	Dektite 0 ~ 50mm	1
19	180HD20	20mm Hot Dip Galvanised Half Saddle	50
20	9020CMS20GY	Corrugated PVC Conduit. 20mm Medium Duty Sheathed	25
21		UV Insulation Tape	2
22		Self-amalgamating Tape	1
23	n/a	6mm2 earth wire (m)	10
24	n/a	6mm2 earth wire lugs	10

Rack			
#	Part Number	Description	Quantity
1	MFB0235904	CABINET SERIES 210 18RU 600MM DEEP	1
2	MFB0272901	CABINET WALLMOUNTING KIT - FOR SERIES 94 (Flat packed)	1
3	MFB0236610	DOOR SERIES 210 18RU ACRYLIC	1
4	MFB0301001	FIXED SHELF - 300D 19IN	1
5	MOLPID00227	PATCH PANEL CAT5E 24 PORT 1U + MGR	1
6	MFB9350040	POWERBOARD HORIZ 2RU 19IN 10 X 10A STD PLUG 2.5M L	1
7	6450 1 050-10	Patch Cord Minder	2

Cable			
#	Part Number	Description	Quantity
1	6451 5 039-05	Cat5E RJ45-RJ45 Ethernet Patch Cable	2

16.2 Category D: - Cabling

Cable			
#	Part Number	Description	Quantity
1	MOLPID00227	PATCH PANEL CAT5E 24 PORT 1U + MGR	1

17 Appendix K – Microwave Antenna PTP 600 Alignment

17.1 Microwave Antenna Alignment Procedure

Before aligning the antennas, ensure that installation on both the Master and Slave sites have been correctly installed (Does not include network installation). **If in doubt recheck the site.**

Prior to starting refer to the configurations tables as listed for each site. Take note of the following:

- Locations IDs between the Master and Slave Buildings
- Bearing between the Master and Slave Buildings
- Antenna Tilt to be used for the site installation

Helpful Hints

- Use a compass to set accurate bearings.
- It is best to use small increments when applying changes to elevation and azimuth.

PART A

STEP 1:

Connect the horizontal / vertical polarization antenna to the ODU if not already done so.

STEP 2: Bearing Alignment

Using a compass set the bearing at each end of the link. Adjust the antenna to point at the other end of the link.

STEP 3: Adjust Elevation and Azimuth Slave Antenna

Adjust the elevation and azimuth of the slave antenna, so as to achieve the highest level of receive signal possible. This is indicated by an increasing numerical value. Do not move the master antenna while doing this task. (Refer to the supplied notes for the Azimuth settings)

NOTE It may be necessary to monitor the received signal strength. Use the ODU installation tones method or the ODU graphical interface install method as below).

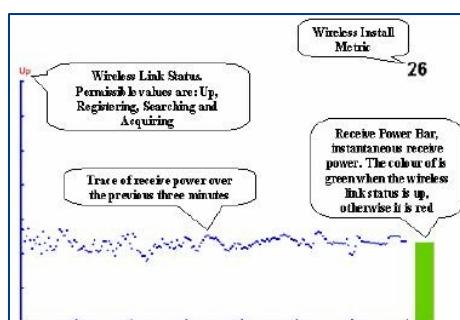


Figure 1 Graphical Interface Install screen

STEP 4: Adjust Elevation and Azimuth Master Antenna

Without moving the Slave antenna, adjust the elevation and azimuth of the Master antenna to achieve the highest receive signal level.

NOTE It may be necessary to monitor the received signal strength. Use the ODU installation tones method or the ODU graphical interface install method as below).

STEP 5: Fine tuning

Repeat steps 3 and 4 as necessary to fine-tune the alignment to find the centre of the beam. The received signal level should increase as the antenna main beams converge to the centre.

STEP 6: Aligned to microwave beam centre

When the antennas have been aligned on the centre of the beam, verify that the receive level is within the predicted range (from the installation report) If this is not the case, go back to step 3

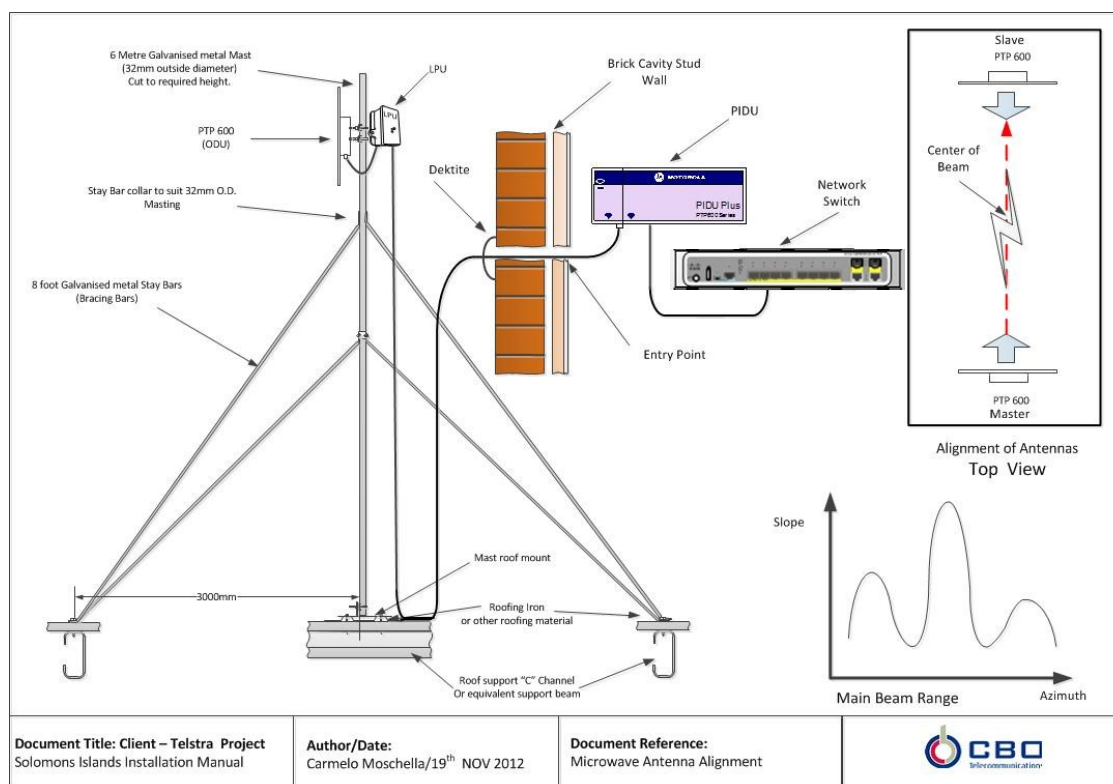


Figure 2 - Installed Antenna

HINT:

The current value of receive level can be verified by using the graphical installation method or by selecting menu option Status and monitoring the Receive Power attribute on the System Status page.



If after repeated attempts to align, the receive level still does not lie within the predicted range, this may be because the data provided to the prediction tool (such as PTP LINK Planner) is inaccurate. Check this data and update the prediction as necessary.

STEP 7: Fine tuning

Once the antennas have been aligned correctly, tighten the connectorised antenna mountings. To ensure that the action of tightening does not alter antenna alignment, continue to monitor received signal level.

PART B

STEP 10: Fine tuning

Re-connect the horizontal polarization antennas.

STEP 11: Waterproofing

Weatherproof the antenna connections at the 'H' and 'V' interfaces of the ODUs, using PVC electrical tape.

18 Appendix L – Tools

This section provides an illustrative guide of hand tools required for installing cabling/mounting Microwave Outdoor Units and setting up Communications Networking Rack Units on site.

On the following page refer to the illustrative guide of tools. Use the check list provided to ensure you have all tools needed for tasks on site.

 **NOTE** Do not go on site unless you have all the tools required.

18.1 Appendix L - Tool Bag

TOOL LIST



Figure 1 Tools Illustrated

Tools Listing

1. Einhell 5pc cordless 18V li-ion Tool kit & Bag
2. Stanley 560g Claw Hammer
3. 300mm Box frame Hacksaw
4. 300mm Hacksaw blades Assorted TPI
5. Bosch 18V Li-Ion Drill driver with 240 Piece Access Package (Not pictured)
6. 6p / 8p Modular crimping tool
7. 110 IDC Impact Punch down tool REF: 43223
8. 111 IDC Impact Punch down tool REF:43222
9. Lan & Coax Cable Tester Ref 43033
10. Drill Set 19Pce 1-10mm
11. Ladder Ext Gorilla 150kg 2.4-3.9
12. Wire Cutters
13. Tinsnips
14. Multitool
15. Tool Kit Supatool 300pce 4 drawer REF:S010030

19 Appendix N – Glossary

The following section provides a definition of any terms, abbreviations or acronyms used within this report.

Acronym/Abbreviation/	Definition
A/C	Air Conditioning
AC	Alternating Current (240 volts, mains supply)
ADSL	Asymmetric Digital Subscriber Line
AP	Access Point (Consists of a radio and an antenna)
ARP	Address Resolution Protocol
AS	Australian Standards
ASA	Adaptive Security Appliances
ATPC	Automatic transmit Power Control
BER	Bit Error Rate
BOM	Build of Materials
BSU	Wi-Max Base Station Unit
C	Celsius
Catenary Wires	Overhead lines or overhead wires are used to transmit electrical energy
CAT5e	Designation for cable type used in networking
CMM or CMM4	Cluster Management Module
CSM	Cluster Scriber Module
CPE	Customer Premise Equipment
dB	Decibel
DC	Direct Current
EWP	Elevated Work Platform (e.g.: Scissor lift, etc.)
FE	Fire Extinguisher
FT	Feet
GHz	Gigahertz
GPO	General Power Outlet
GPS	Global Positioning System
HTP	High Through Point
IDU	Indoor Unit

KM	Kilometres
L2	Layer 2; the Data layer of the OSI Model
L3	Layer 3; the Network layer of the OSI Model
LAN	Local Area Network
LOS	Line Of Sight
LPU	Lightning Protection Unit
LPU	Lightning Protection Unit
M	Metres
Mbps	Megabits per second
MHz	Megahertz
MM	Millimetres
NEMS	Network Environment Monitoring System
ODU	Outdoor Unit
PDU	Power Distribution Unit
PMP	Point-to-Multipoint
PTP	Point To Point
RCD	Residual Current Device
RF	Radio Frequency
RU	Rack Unit
SM	Subscription Manager
TER	Telecommunications Equipment Room ¹
UPS	Uninterrupted Power Supply
UV	Ultra Violet
WAN	Wide Area Network
WIC	WAN Interface Card
WiMAX	Worldwide interoperability for microwave access

¹ A Telecommunications Equipment Room is a stand-alone communications room generally associated with a remote tower site. These rooms may contain a combination of server and communication equipment.

20 Appendix O -Certificate of Acceptance

This document (referred to as the 'Installation Manual') and all listed appendices constitute the documentation record of the project scope of works outlined in the document:

1562 Solomon Islands Government–Metropolitan Area (Wireless) Network Installation / Support Manual for Solomon Islands Government Issue 1_v1_0.pdf

This document is the conclusion of works related to the above project number. The acceptance of all deliverables is considered final without receipt of objection within seven days.

Signed on behalf of (Client) by a duly authorised representative:

Name (printed):

Date:
